

Problem 3. (Eigendecomposition Algorithms, 25 points)

- 1) Construct a square matrix with eigenvalues $\{1,2,3,4,5\}$, and then use the power method to verify the five eigenvalues. Record the error for each iteration and plot the convergence curve (i.e. error between each eigenvalue generated by the power method and the corresponding true eigenvalue versus iteration). Matlab code is also required. (Note: Randomly generate an orthogonal matrix $\mathbf{X}$ and let $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$, where $\mathbf{\Lambda} = \text{Diag}([1, 2, 3, 4, 5])$. You can design your own stopping rules; it can be the maximum number of iterations or convergence of eigenvalues) (7 points)
- 2) Use QR iteration with Rayleigh quotient shift to find an eigenvalue of the matrix constructed with eigenvalues $\{1,5,9,11,14\}$. Plot the convergence curve and compare it with the curve generated by QR iteration without shift. (Note: You need to plot two curves in one figure.) Matlab code is also required. (8 points)
- 3) Consider the following **Algebraic Riccati Equation** (ARE) in its generalized symmetric form:

$$G + XA + A^T X - XFX = 0, \quad (1)$$

This equation arises frequently in optimal control (e.g., LQR design), estimation (e.g., Kalman filtering), and differential games. Provide a Matlab function `solve_are_general(A, G, F)` that computes the unique stabilizing symmetric solution $X = X^T$ to Equation (1) and plot the distribution of eigenvalues of $A - FX$ and the Hamiltonian matrix $\begin{bmatrix} A & -F \\ -G & A^T \end{bmatrix}$. The data are given as follows:

- A: an $n \times n$ real matrix $\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$,
- G: an $n \times n$ symmetric positive semi-definite matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$,
- F: an $n \times n$ symmetric positive semi-definite matrix $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$.

Requirements:

- You **may not** use built-in solvers including MATLAB's `care`, `icare`, etc.
- Your implementation must be based on a numerical method (e.g., Schur decomposition of a Hamiltonian matrix, Newton iteration, or invariant subspace method).
- The solution X must be real and symmetric.

(Note: One may begin by constructing the Hamiltonian matrix and computing its Schur decomposition. The eigenvalues can then be reordered using the Matlab `ordschur` function. Finally, the reordered symplectic and triangular matrices can be employed to compute the desired solution.)(10 points)

Solution: